

Experiment 5

Module Name: Automation with Selenium

Experiment Title: The HTML Web Table Extractor

Problem Statement: Extract a specific data point from an HTML table by matching a value in another column.

Objective: Iterate through <tr> and <td> tags to dynamically retrieve corresponding row data.

Tools, software, and concepts used: Python, Selenium, XPath axes, multi-element iteration.

Source code or implementation steps:

```
from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()

try:
    driver.get("https://the-internet.herokuapp.com/tables")

    # Target: Table 1. Find the 'Due' amount for user 'Doe, Jason'
    rows = driver.find_elements(By.XPATH, "//table[@id='table1']/tbody/tr")

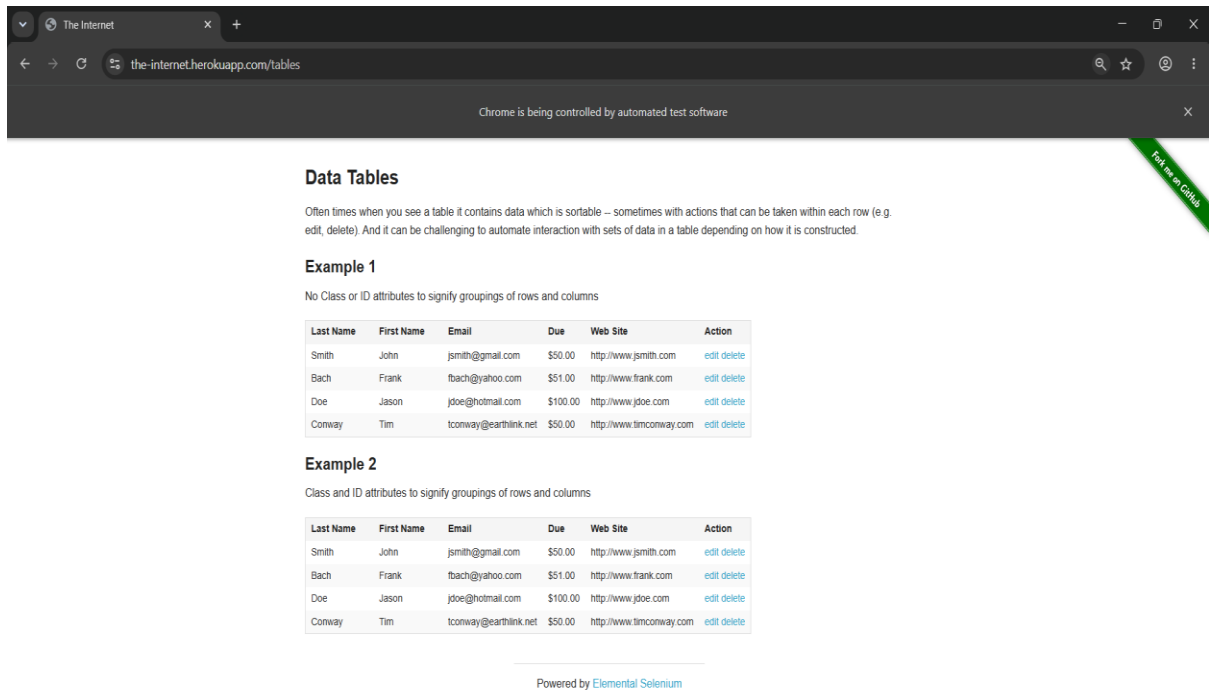
    target_amount = ""
    for row in rows:
        # Get all columns (td) for the current row
        cols = row.find_elements(By.TAG_NAME, "td")
        last_name = cols[0].text
        first_name = cols[1].text

        if last_name == "Doe" and first_name == "Jason":
            target_amount = cols[3].text # The 'Due' column
            break

    print(f"Jason Doe owes: {target_amount}")
    assert target_amount == "$100.00"

finally:
    driver.quit()
```

Output with clearly labelled screenshots:



Data Tables

Often times when you see a table it contains data which is sortable -- sometimes with actions that can be taken within each row (e.g. edit, delete). And it can be challenging to automate interaction with sets of data in a table depending on how it is constructed.

Example 1

No Class or ID attributes to signify groupings of rows and columns

Last Name	First Name	Email	Due	Web Site	Action
Smith	John	jsmith@gmail.com	\$50.00	http://www.jsmith.com	edit delete
Bach	Frank	fbach@yahoo.com	\$51.00	http://www.frank.com	edit delete
Doe	Jason	jdoe@hotmail.com	\$100.00	http://www.jdoe.com	edit delete
Conway	Tim	tconway@earthlink.net	\$50.00	http://www.timconway.com	edit delete

Example 2

Class and ID attributes to signify groupings of rows and columns

Last Name	First Name	Email	Due	Web Site	Action
Smith	John	jsmith@gmail.com	\$50.00	http://www.jsmith.com	edit delete
Bach	Frank	fbach@yahoo.com	\$51.00	http://www.frank.com	edit delete
Doe	Jason	jdoe@hotmail.com	\$100.00	http://www.jdoe.com	edit delete
Conway	Tim	tconway@earthlink.net	\$50.00	http://www.timconway.com	edit delete

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Screenshot 1: HTML table displaying user data

Console Output: Jason Doe owes: \$100.00

Brief result: Correctly isolated the target cell based on the text context of its sibling cells.

Observation: By gathering all rows into a list and nesting a column search, the script remains resilient even if the row order changes dynamically on the server.

Conclusion: Logical iteration through DOM structures allows for reliable data scraping from tables.